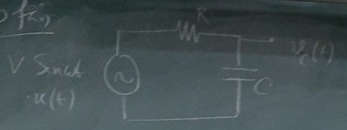
① So far,



Take the s-domain route

$$\to \frac{A_1}{s-j\omega} + \frac{A_2}{s+j\omega} + \frac{A_3}{s+\frac{1}{RC}}$$

$$\underbrace{\frac{A_1}{s-j\omega} + \frac{A_2}{s+j\omega}}_{*}$$

It turns out $A_1 = A_2^*$

$\hookrightarrow * = B_1\cos\omega t + B_2\sin\omega t$

We can use this & skip s-domain analysis

directly write $v_c(t) \to B_1\cos\omega t + B_2\sin\omega t + B_3 e^{-t/RC}$

use initial condns to get B_1, B_2, B_3

Suppose $v_c(0_-) = 0$

initial condn $\Rightarrow$ $B_1 + B_3 = 0$

One more condn

$$C\frac{dv_c}{dt} = \frac{V\sin\omega t - v_c(t)}{R}$$

$\Updownarrow$

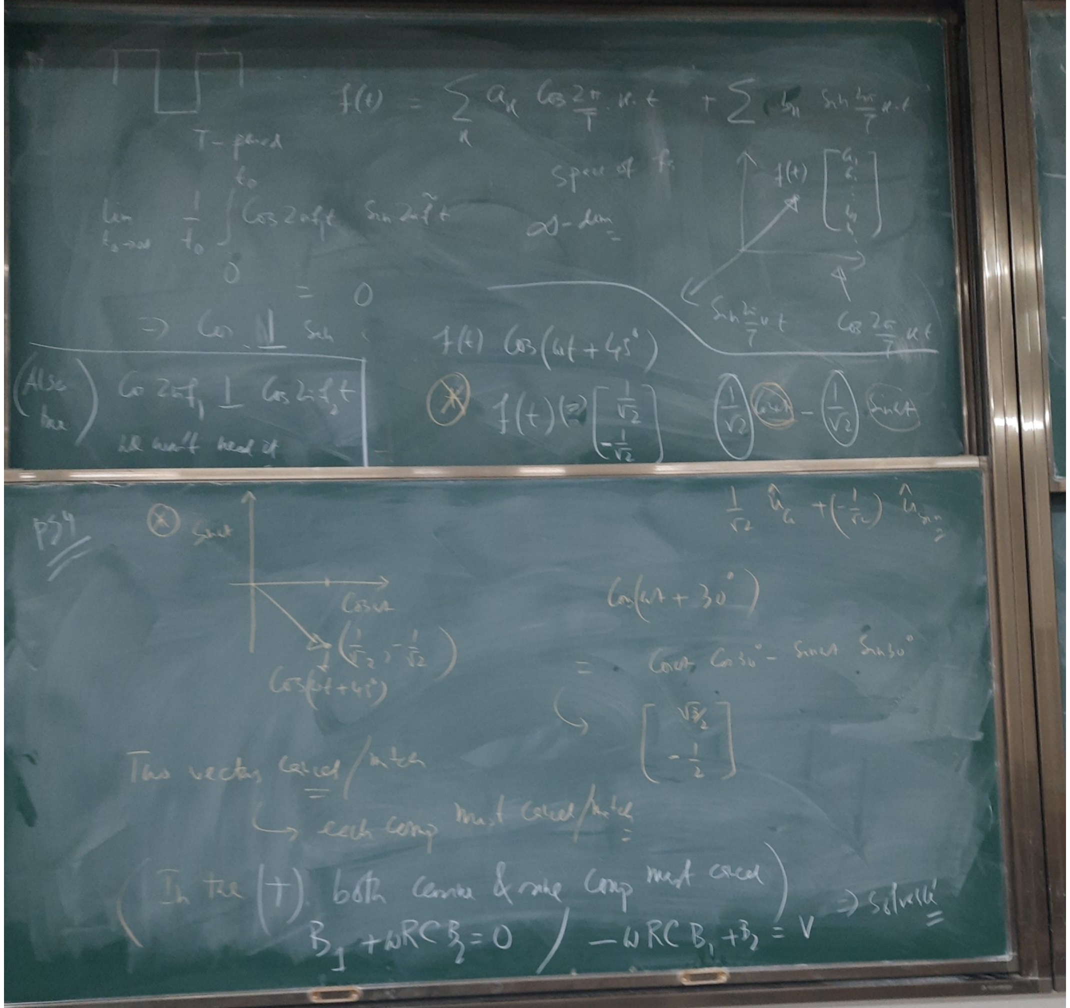
(Another way $RC\frac{dv_c}{dt} + v_c = v_s$)

$(e^{-t/RC} \to 0) \Leftarrow t \to \infty$ ((B_1, B_2 matter)

but $B_3 \neq 0$

(†) $RC(-\omega B_1\sin\omega t + \omega B_2\cos\omega t) + B_1\cos\omega t + B_2\sin\omega t = V\sin\omega t$ $\boxed{\text{for all } t}$

$\Leftrightarrow$ "Cos & sine terms must match separately"



$$f(t) = \sum_n a_n \cos\frac{2\pi}{T}\cdot n\cdot t + \sum b_n \sin\frac{2\pi}{T} n\cdot t$$

T-period

$$\lim_{t_0\to\infty} \frac{1}{t_0}\int^{t_0} \cos 2\pi f_1 t \; \sin 2\pi \tilde{f} t$$

0

$= 0$

$\Rightarrow$ Cos $\perp$ Sin

(Also true) $\cos 2\pi f_1 \perp \cos 2\pi f_2 t$

We won't need it

Space of fs

∞-dim

$$f(t) \quad \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ a_n \\ b_n \end{bmatrix}$$

$\sin\frac{2\pi}{T} n t$ $\qquad$ $\cos\frac{2\pi}{T} n t$

$f(t)$ $\cos(\omega t + 45^\circ)$

⊛ $f(t) \Leftrightarrow \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ $\quad \frac{1}{\sqrt{2}} \cos\omega t - \frac{1}{\sqrt{2}} \sin\omega t$

$$\frac{1}{\sqrt{2}} \hat{u}_c + \left(-\frac{1}{\sqrt{2}}\right) \hat{u}_{\sin}$$

PS4

⊛ $\sin\omega t$ $\quad \cos\omega t$ $\quad \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ $\quad \cos(\omega t + 45^\circ)$

$$\cos(\omega t + 30^\circ)$$

$$= \cos\omega t \cos 30^\circ - \sin\omega t \sin 30^\circ$$

$$\hookrightarrow \begin{bmatrix} \sqrt{3}/2 \\ -\frac{1}{2} \end{bmatrix}$$

Two vectors cancel/match

↪ each comp must cancel/match

(In the (†) both cosine & sine comp must cancel)

$B_1 + \omega RC B_2 = 0$ / $-\omega RC B_1 + B_2 = V$ $\Rightarrow$ Solvable once B_1, B_2 known plg in $\underline{B_1 + B_3 = 0}$ $\Rightarrow$ $B_3 = V$

$$V_c(t) = \frac{-\omega RC}{1+\omega^2R^2C^2}\cos\omega t + \frac{1}{1+\omega^2R^2C^2}\sin\omega t + \frac{\omega RC}{1+\omega^2R^2C^2}\, e^{-t/RC}$$

At $\underline{t \to \infty}$ $\quad V_c(t) = \quad + \quad \underline{0}$ $\quad \to 0 \;\; \underline{t \to \infty}$

Suppose we are only interested in $\underline{t \to \infty}$, only $(\dagger)$ matters!

$$V_c(t) = \bar{k}\,\sin(\omega t - \theta)$$

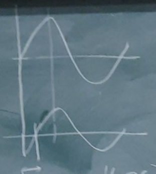
$$\theta = \tan^{-1}(\omega \underline{RC})$$

freq $\quad \tau$

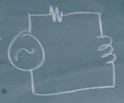
phase diff $= \omega \cdot (\text{time const})$

PS5

PS6



$\theta = \tan^{-1}(\omega RC)$



$$\theta = \omega \cdot \frac{L}{R}$$

Are we convinced? if c/f $V_0\cos(\omega t+\phi)$ $(A_1\cos + A_2\sin)$

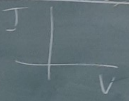
(Because all C/L/R is multiply, take $\frac{d}{dt}$ or $\int$) ⇒ any curr./volt ⇒ $K\cos(\omega t+\phi)$

sin/cos → sin/cos

$(K_1\cos + K_2\sin)$

(at $\boxed{t\to\infty}$) only

If we understand



char. of R/L/C for V/I

$A_1\cos + A_2\sin$

⇒ We know simple KCL/KVL

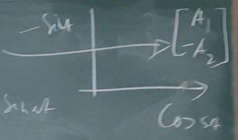
(p57)

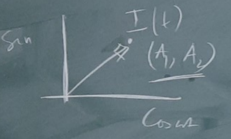
(p58)

Resistance

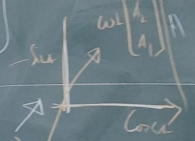
$I(t) = A_1\cos\omega t + (-A_2)(-\sin\omega t)$

$\to V(t) = R\,A_1\cos\omega t + R\,A_2\sin\omega t$





$$\begin{pmatrix}A_1\\A_2\end{pmatrix}\to R\begin{pmatrix}A_1\\A_2\end{pmatrix}$$



"NOT a rotation"

(NOT very human friendly

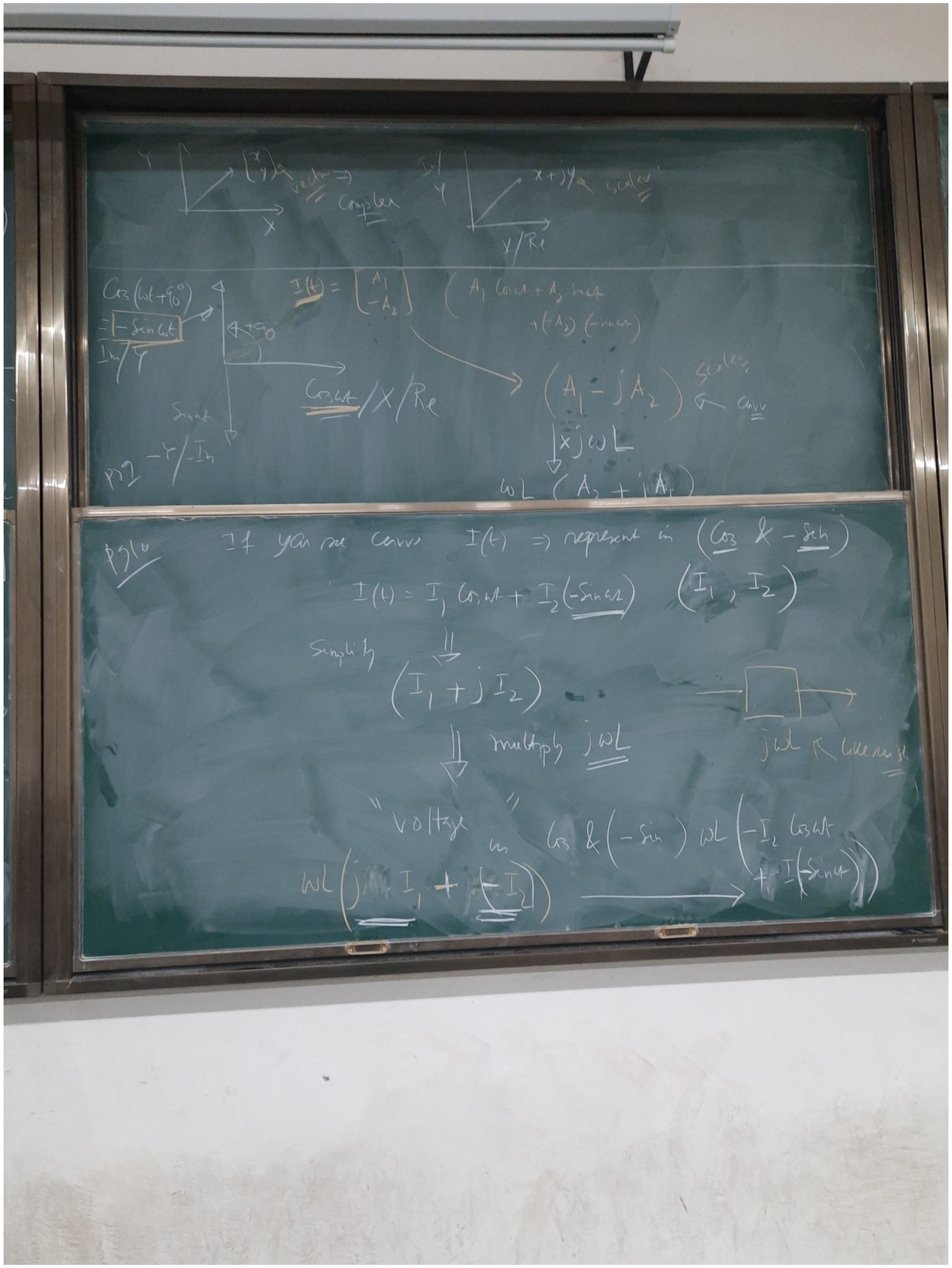
Inductor I(t) same

$$V(t) = L\frac{dI(t)}{dt} \Rightarrow V(t) = \omega L\left(A_1(-\sin\omega t) + A_2\cos\omega t\right)$$

$$\begin{bmatrix}A_1\\-A_2\end{bmatrix}\to\omega L\begin{bmatrix}+A_2\\+A_1\end{bmatrix}$$

$$\begin{matrix}A_1\\A_2\end{matrix}\to\omega L\begin{bmatrix}0&1\\-1&0\end{bmatrix}\begin{bmatrix}A_1\\A_2\end{bmatrix}$$

iii.



$\begin{pmatrix} x \\ y \end{pmatrix}$ vector ⇒ Complex

$x + jy$ scalar

$\cos(\omega t + 90^\circ)$

$= \boxed{-\sin\omega t}$

Im / Y

$+90^\circ$

$\cos\omega t$ / X / Re

$\sin\omega t$

pg 9 $-Y / -Im$

$I(t) = \begin{bmatrix} A_1 \\ -A_2 \end{bmatrix}$ $\Big(A_1 \cos\omega t + A_2 \sin\omega t$

$+ (-A_2)(-\sin\omega t) \Big)$

$(A_1 - jA_2)$ scalar ← conv

$\times j\omega L$

$\omega L\ (A_2 + jA_1)$

pg 10 If you see conv $I(t)$ ⇒ represent in $(\cos \ \& \ -\sin)$

$I(t) = I_1 \cos\omega t + I_2(-\sin\omega t)$ (I_1, I_2)

Simplify ⇓

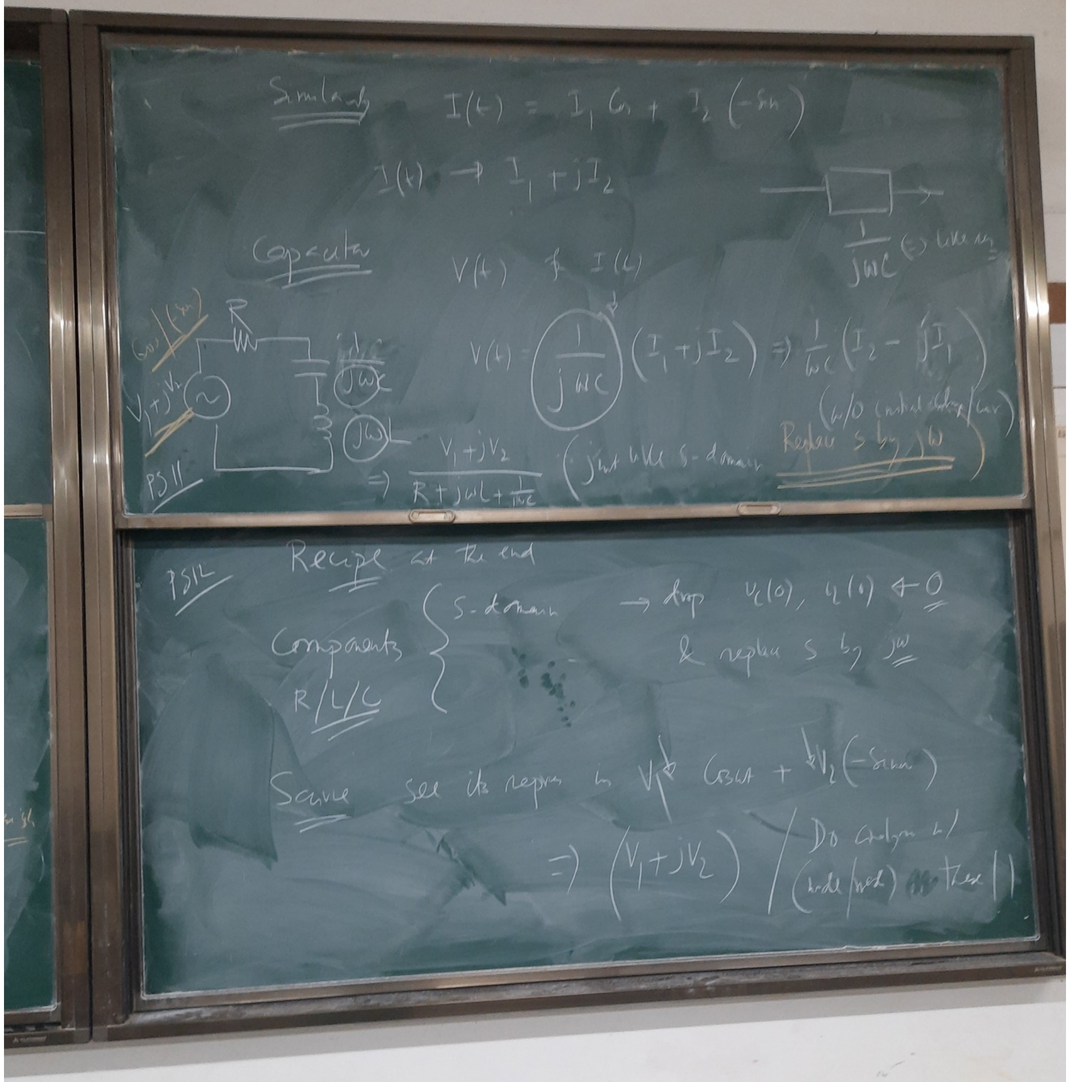
$(I_1 + jI_2)$

⇓ multiply $j\omega L$

$j\omega L$ ↖ like res..

"voltage" in $\cos \ \& \ (-\sin)$ $\omega L \Big(-I_2 \cos\omega t + I_1(-\sin\omega t) \Big)$

$\omega L \big(jI_1 + (-I_2) \big)$ ⟶



Similarly $I(t) = I_1 \cos + I_2 (-\sin)$

$I(t) \rightarrow I_1 + jI_2$

$\frac{1}{j\omega C}$ ⟺ like R

Capacitor $V(t)$ & $I(t)$

$V(t) = \boxed{\frac{1}{j\omega C}} (I_1 + jI_2) \Rightarrow \frac{1}{\omega C}(I_2 - jI_1)$

(w/o initial charge/curr)

Cos / (-sin)
R
$\frac{1}{j\omega C}$
$j\omega C$
$j\omega L$
$V_1 + jV_2$
PS 11

$\Rightarrow \frac{V_1 + jV_2}{R + j\omega L + \frac{1}{j\omega C}}$ (just like s-domain Replace s by $j\omega$)

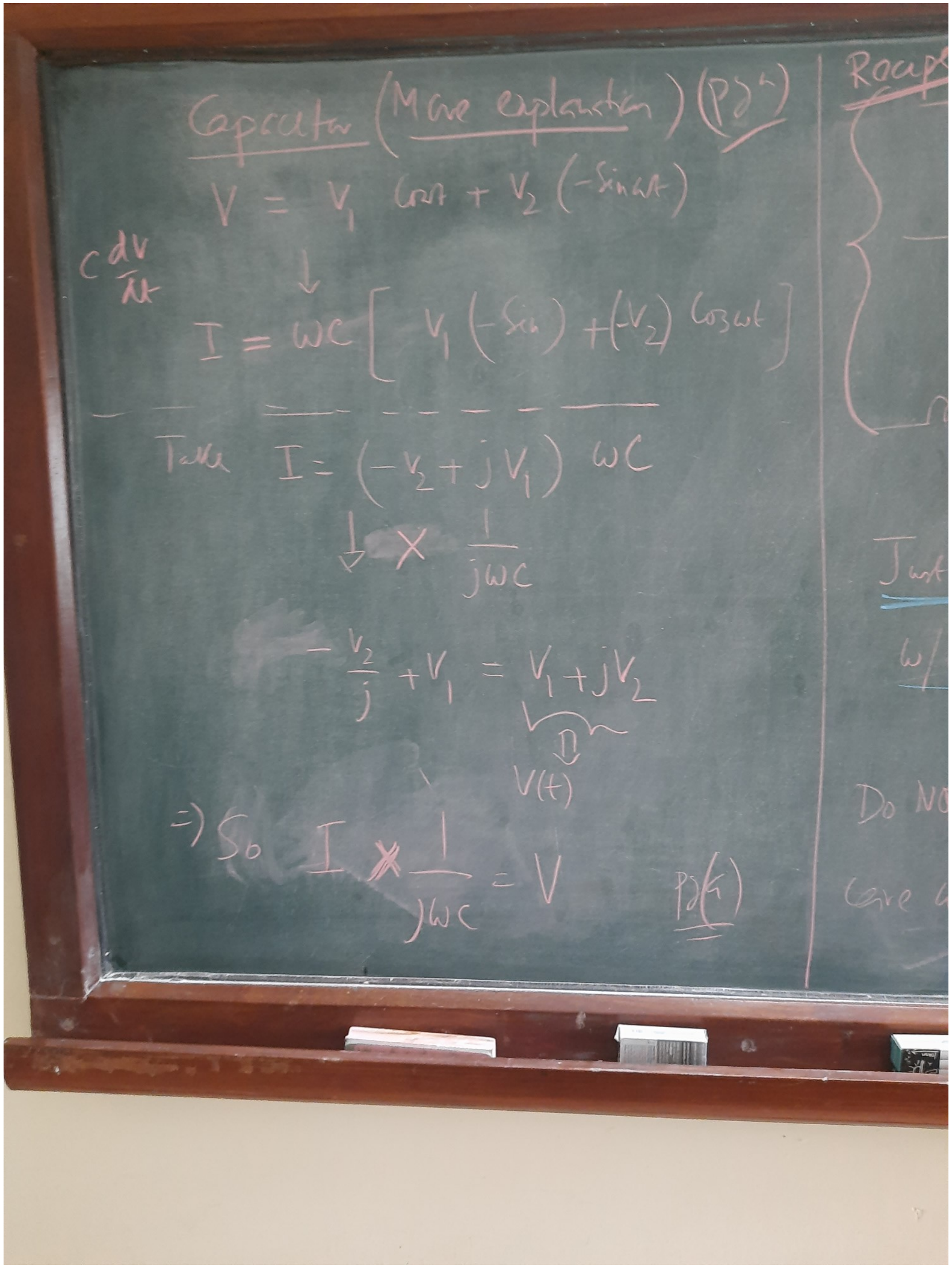
PS12

Recipe at the end

Components R/L/C { s-domain → drop $v_C(0)$, $i_L(0)$ ← 0 & replace s by $j\omega$

Source see its repres in $V_1 \cos\omega t + V_2(-\sin\omega t)$

$\Rightarrow (V_1 + jV_2)$ / Do analysis w/ (node/mesh) on these!!



Capacitor (More explanation) (Pg^{n})

$$V = V_1 \cos\omega t + V_2(-\sin\omega t)$$

$$\Big\downarrow C\frac{dV}{dt}$$

$$I = \omega C\left[V_1(-\sin) + (-V_2)\cos\omega t\right]$$

Take $I = (-V_2 + jV_1)\,\omega C$

$$\Big\downarrow \times \frac{1}{j\omega C}$$

$$-\frac{V_2}{j} + V_1 = \underbrace{V_1 + jV_2}_{\Downarrow\; V(t)}$$

$\Rightarrow$ So $I \times \frac{1}{j\omega C} = V$ Pg(n)

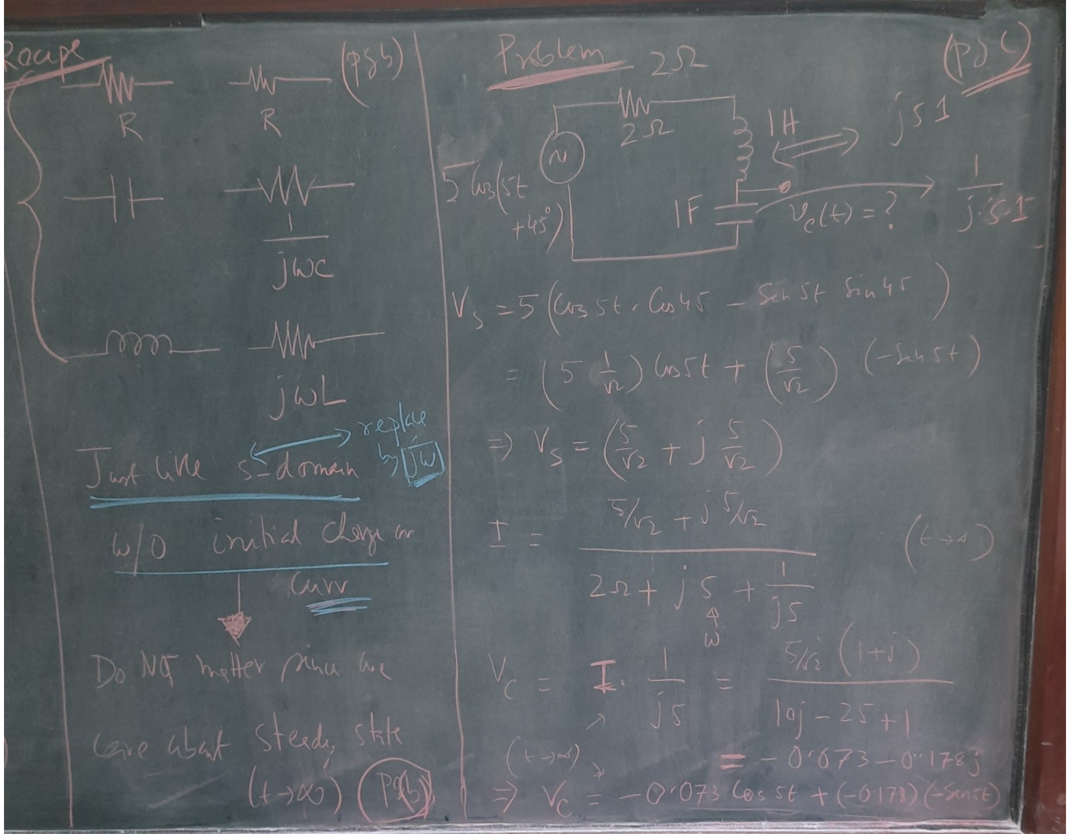
Recap

Just

w/

Do NO

Give



Recap (P8b)

$R \to R$

$\dashv\vdash \to \frac{1}{j\omega C}$

$\text{ⅿⅿ} \to j\omega L$

Just like s-domain $\xleftrightarrow{}$ replace s by $\boxed{j\omega}$

w/o initial charge or curr

Do NOT matter since we care about steady state $(t \to \infty)$ (P8b)

Problem (P8c)

2Ω
2Ω
5 cos(5t +45°)
1H
1F

$1H \Longleftrightarrow j\,5\,1$

$1F \longrightarrow \frac{1}{j \cdot 5 \cdot 1}$

$v_c(t) = ?$

$$V_s = 5(\cos 5t \cdot \cos 45 - \sin 5t \sin 45)$$

$$= \left(5 \tfrac{1}{\sqrt2}\right)\cos 5t + \left(\tfrac{5}{\sqrt2}\right)(-\sin 5t)$$

$$\Rightarrow V_s = \left(\tfrac{5}{\sqrt2} + j\tfrac{5}{\sqrt2}\right)$$

$$I = \frac{5/\sqrt2 + j\,5/\sqrt2}{2\Omega + j\underset{\omega}{\underset{\uparrow}{5}} + \frac{1}{j5}} \qquad (t \to \infty)$$

$$V_c = I \cdot \frac{1}{j5} = \frac{5/\sqrt2\,(1+j)}{10j - 25 + 1}$$

$$(t \to \infty) \qquad = -0.073 - 0.178j$$

$$\Rightarrow V_c = -0.073\cos 5t + (-0.178)(-\sin 5t)$$